known as the “competitive exclusion principle,” was set out by Russian microbiologist Georgy Gause and is also known as Gause’s Law.

Gause devised his principle from laboratory experiments, using cultures of microorganisms, rather than from observations in nature. In nature, he proposed, there were too many variables to draw conclusions about how ecological mechanisms work. He argued that little progress had been made since Darwin’s era in understanding how species compete for survival, whereas the experimental method had produced great advances in areas such as genetics. In fact, the competitive exclusion principle—although a useful theoretical model—is rarely seen in nature, simply because, in a bid to survive, a weaker competitor tends to quickly move on or adapt.

Avoiding competition

Most creatures can make the changes necessary for survival. A variety of birds can live in a garden during any one year because they all operate in different “niches.” They have contrasting beak shapes and sizes that allow them to eat different types of food—the robin preferring insects, the finch eating seeds. Their choice of habitat and feeding times might also vary; this is known as resource partitioning.

In 1957, Robert MacArthur noted this phenomenon in North American warblers. The five species he observed, each with distinctive, colorful markings, flitted in and out of coniferous trees, feeding on bugs and other insects. They could coexist in one habitat because they did not try to feed in the same part of the tree but at different heights and depths of the foliage. In this way they avoid competing with each other.